



Different Ways to Access LLMs (Step-by-Step Guide)

Objective

To understand different ways of accessing Large Language Models (LLMs) — including

- **Web-based Access**
 - ChatGPT
 - Claude
 - OpenAI Playground
 - Claude Console
 - Google AI Studio
 - OpenRouter Playground
 - **API based**
 - Groq API
 - Google Gemini API
 - **Integrated APIs / Frameworks**
 - Gemini within Google Colab Notebook
 - **Self-hosted / Local deployments**
 - LMStudio
-

When to use these different ways of accessing LLMs

1. Web-based Access (Playground / Chat Interface)

Examples: ChatGPT (chat.openai.com), Claude (claude.ai), OpenRouter Playground

When to use:

- For **quick testing** of prompts and observing natural chat-style responses.
- To analyze **content quality, coherence, and relevance** without worrying about code or API setup.
- Best suited for **beginners** or **manual qualitative comparisons**.

2. API-Based Access (via Colab or Scripts)

Note: Refer to FAST API reading material for introduction to APIs



Examples: Groq API to use OpenAI and Llama models, Google Gemini API

When to use:

- For **programmable benchmarking** — when you need to capture metrics like **latency**, **tokens/sec**, and **response length** automatically.
- To integrate LLMs into **applications or evaluation pipelines**.
- Ideal for **developers and data scientists** performing automated comparisons.

3. Integrated Access (within Applications / Frameworks)

Examples: Gemini integrated in Colab, LangChain pipelines, or any embedded LLM workflow

When to use:

- When testing LLMs as part of a **larger workflow** (e.g., data querying, summarization, or RAG pipelines).
- To observe how **model reasoning and context handling** change when integrated into an end-to-end application.
- Useful for **evaluating task-specific performance** (not just text generation).

4. Local Model Access

Examples: LM Studio, Ollama, GPT4All

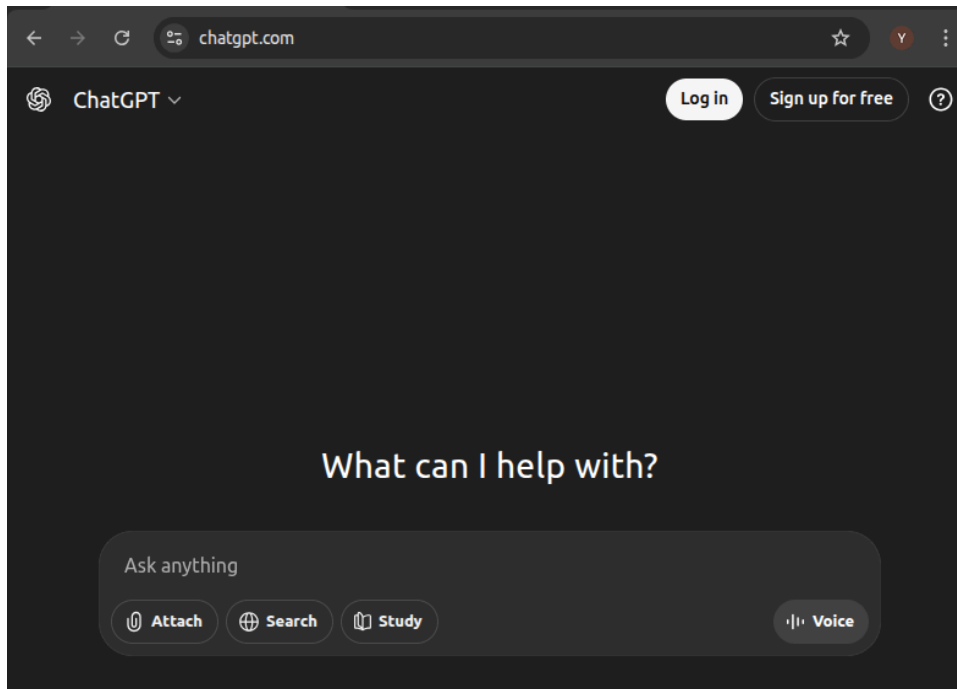
When to use:

- When you want **full offline control** and **data privacy**.
- To compare **response quality and speed** of local models against cloud-hosted APIs.
- Best for exploring **cost-efficiency** and **resource utilization** on your local machine.

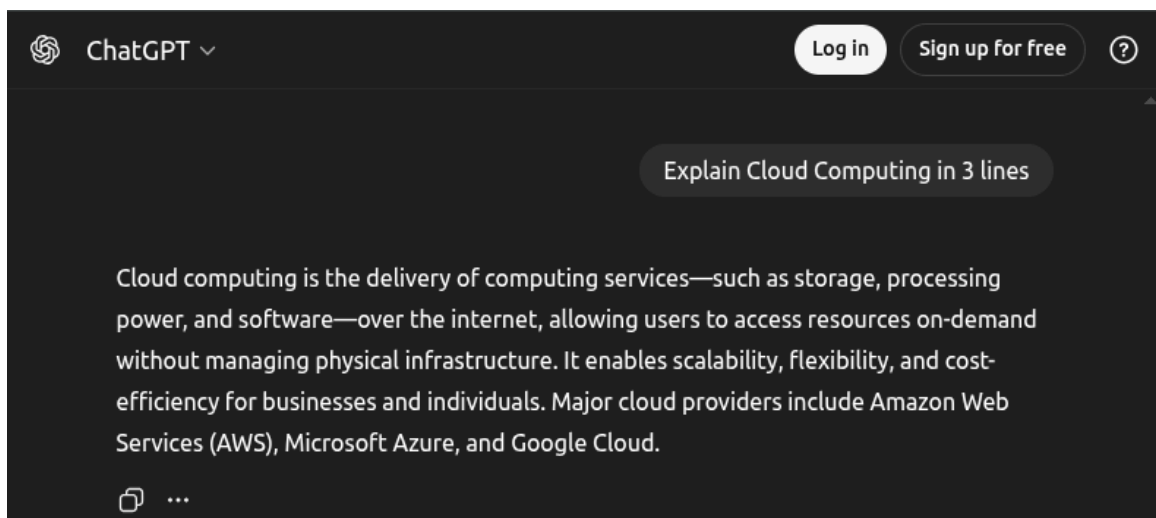
I. Using LLM via Web-based Access

1. ChatGPT (Model: GPT-5)

1. Go to <https://chatgpt.com/>

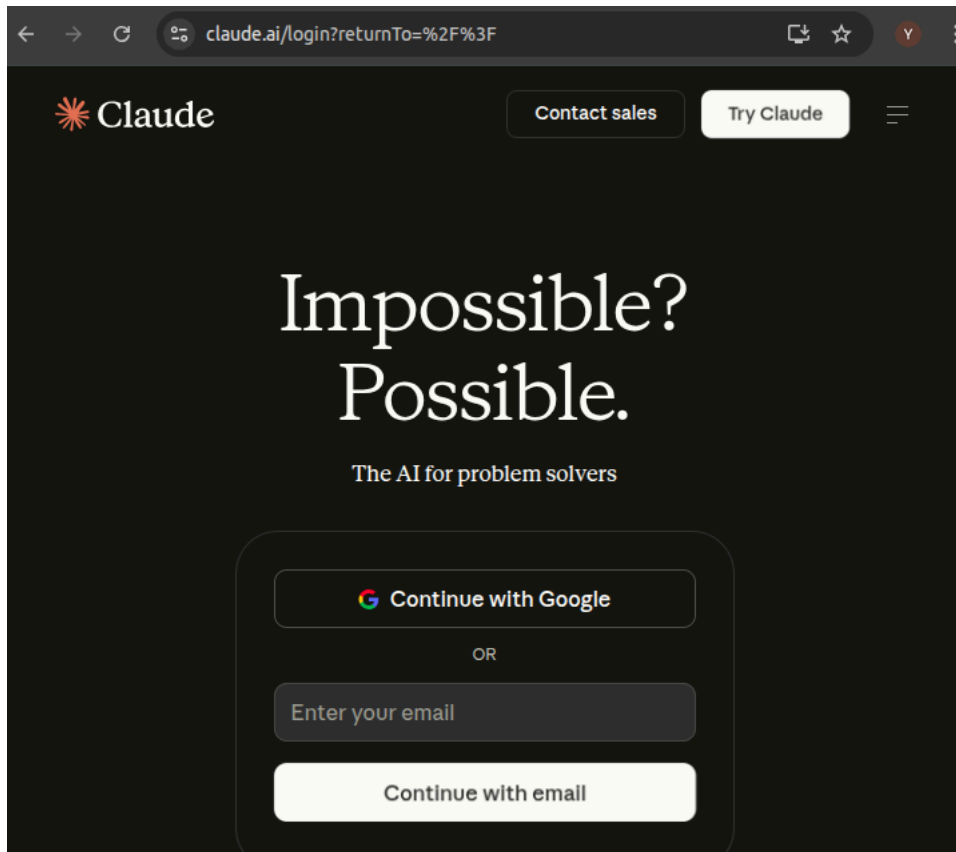


2. Sign up for free
3. Provide your prompt and get the response



2. Claude (Model: Sonnet 4.5)

1. Go to <https://claude.ai/new>



2. Select *Continue with Google*, or *Continue with email*
3. Select *Use Claude for free*

Plans that grow with you



Free

Meet Claude

USD 0

Use Claude for free

- ✓ Chat with Claude on web, iOS, and Android
- ✓ Write, edit, and create content
- ✓ Analyze text and upload images
- ✓ Generate code and visualize data
- ✓ Get web search results inside chat



Pro

Research, code, and organize

Monthly **Yearly - Save 17%**

USD 17

/ month billed annually

Get Pro plan

Everything in Free and:

- ✓ More usage than Free*
- ✓ Access more Claude models
- ✓ Unlimited Projects to organize chats
- ✓ Unlock deep Research tools
- ✓ Use extended thinking for complex work
- ✓ Connect Google Workspace: email, calendar, and docs
- ✓ Integrations with remote MCP
- ✦ Includes Claude Code



Max

Higher limits, priority access

From USD 100

/ month billed monthly

Get Max plan

Everything in Pro, plus:

- ✓ Choose 5x or 20x more usage than Pro*
- ✓ Higher output limits for all tasks
- ✓ Early access to advanced Claude features
- ✓ Priority access at high traffic times
- ✦ Includes Claude Code

4. It might ask to **Install Claude Code**, for now you can select **Maybe later**

NEW

Collab with Claude in your codebase

- Get AI pair programming right in your terminal with Claude Code
- Ship better code faster, and understand complex codebases with ease
- Available with a Claude Pro or Max plan, or billed through your Anthropic Console account based on API usage

Install Claude Code

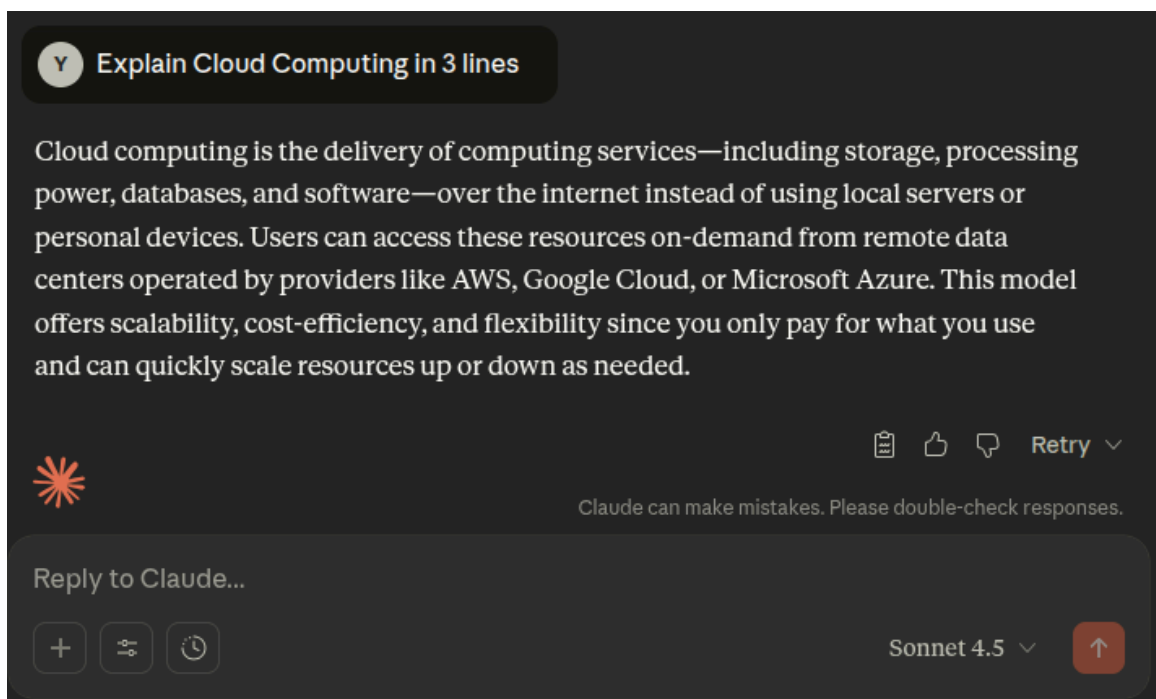
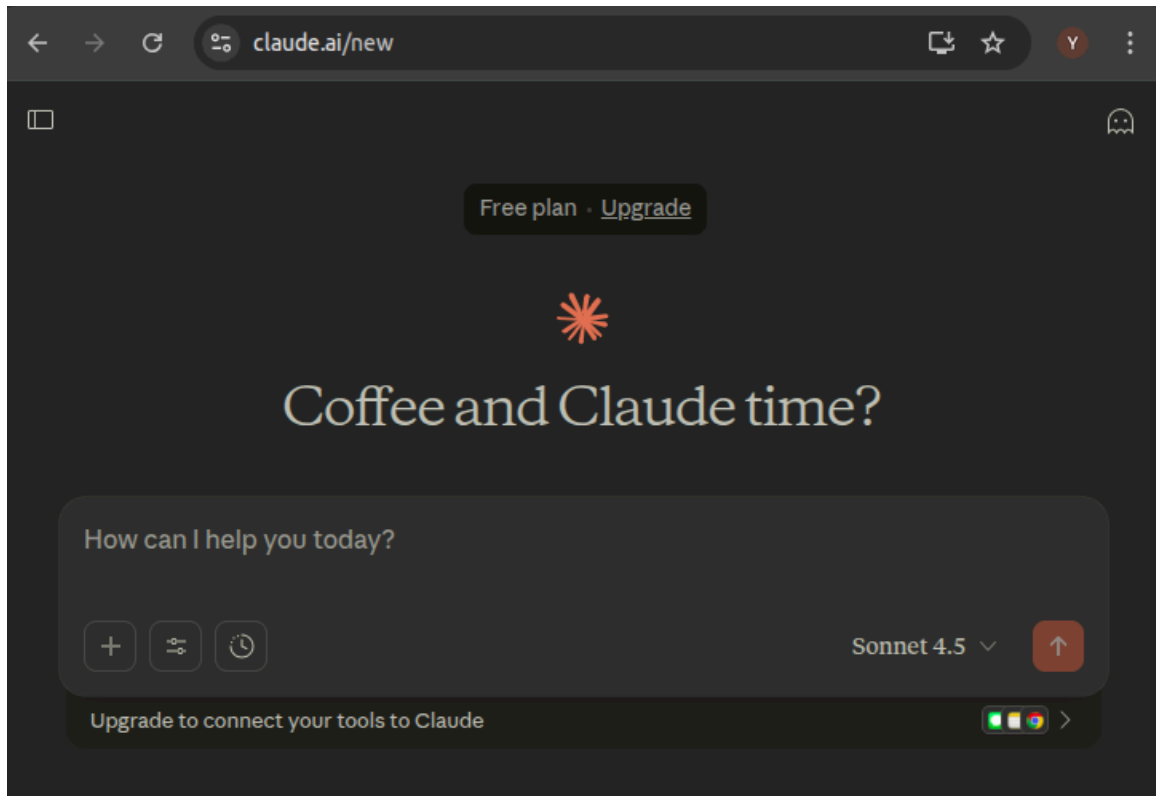
Maybe later

Welcome to the Claude Code research preview!

CLAUDE CODE

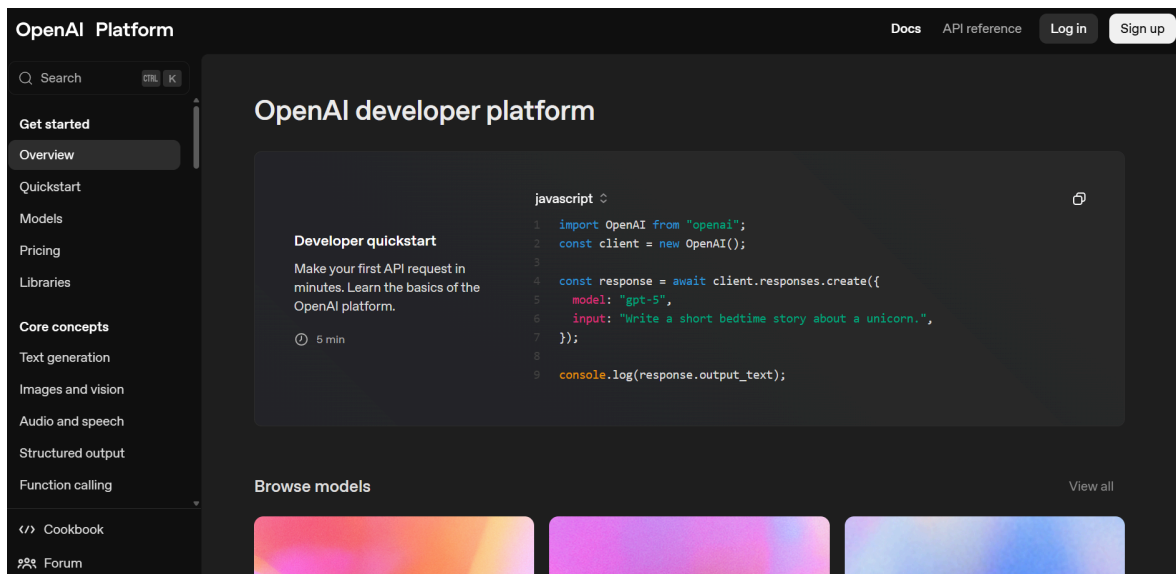
Login successful. Press Enter to continue

5. Provide your prompt and get the response

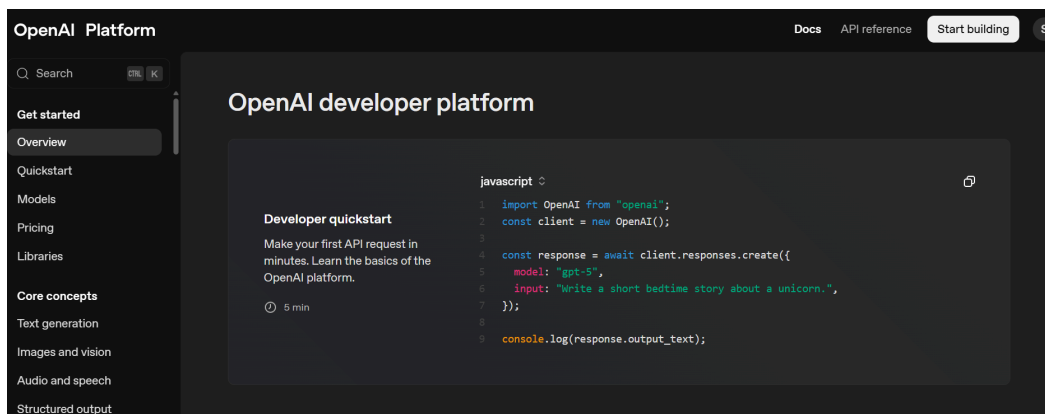


3. OpenAI Playground (GPT Models)

1. Go to <https://platform.openai.com/>

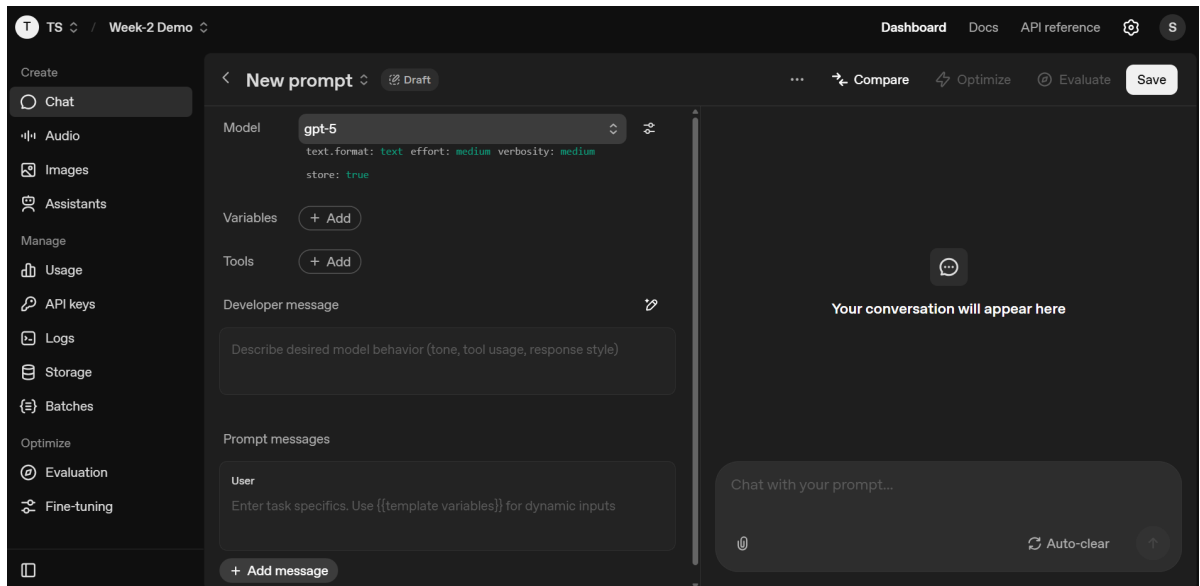


2. Click on **Sign Up** if you don't have an account in OpenAI.
3. After signing up, log in to the account using the credentials.

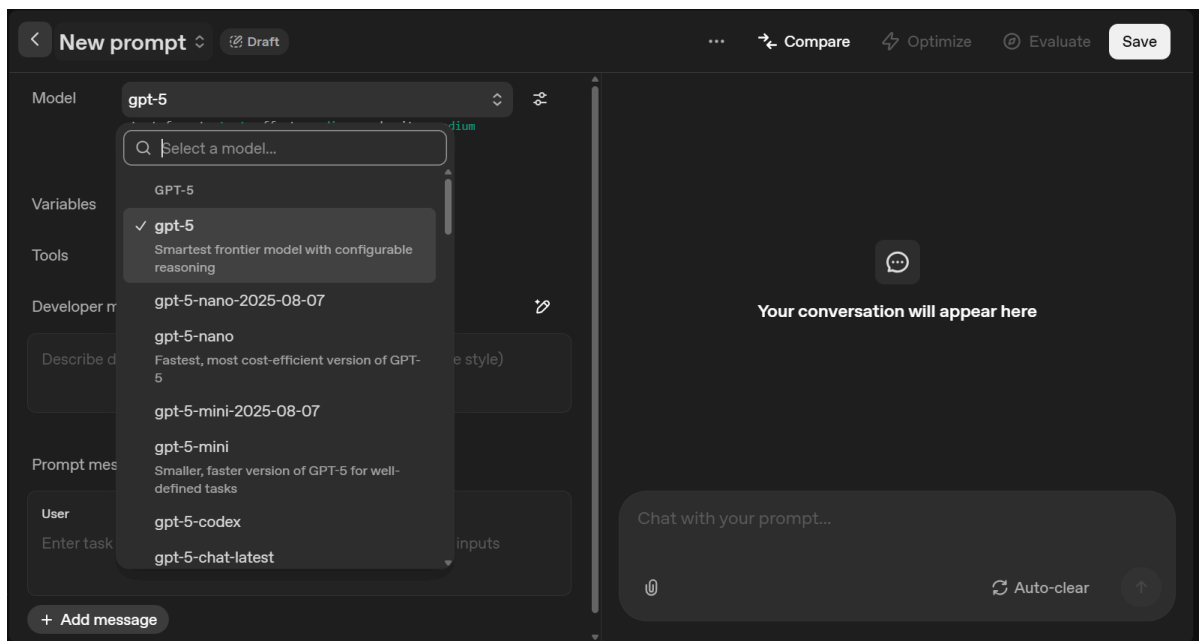


4. Further, access the API playground(Chat) of OpenAI: <https://platform.openai.com/chat>
You will require to create an **Organization** and a **Project**.

5. After Creating the Organization and Project you will see the dashboard as following:



6. Find **Model dropdown** (middle-right) → choose GPT-5 or GPT-4 or any other model that you want for prompting.



7. Paste the **sample prompt** given below in the chat box saying *Chat with your prompt...*

Prompt:



Write a detailed but concise essay (400–500 words) on the topic:

“The role of artificial intelligence in addressing global climate change.”

Your essay should include:

1. A short introduction (2–3 sentences)
2. 2–3 key arguments supported by real-world examples
3. A brief conclusion summarizing your viewpoint

Make sure the content is:

- Clear and logically structured
- Factually correct
- Engaging and easy to follow

Self-Evaluation Section

After completing your essay, provide a **performance report** in **JSON** format like this:

```
```json
{
 "response_length_tokens": <approximate number of tokens in your response>,
 "time_taken_seconds": <estimated total time to generate output>,
 "tokens_per_second": <calculated or estimated speed>,
 "content_quality_score": <1-10>,
 "factual_accuracy_score": <1-10>,
 "relevance_score": <1-10>,
 "coherence_score": <1-10>,
 "creativity_score": <1-10>,
 "overall_confidence_level": "<low|medium|high>"
}
```
```

If you cannot measure speed directly, estimate it based on your normal processing.

New prompt Draft Compare Optimize Evaluate Save

Model: **gpt-5**
text.format: **text** effort: **medium** verbosity: **medium**
store: **true**

Variables: + Add

Tools: + Add

Developer message: Describe desired model behavior (tone, tool usage, response style)

Prompt messages:

User: Enter task specifics. Use {{template variables}} for dynamic inputs

+ Add message

Your conversation will appear here

Write a detailed but concise essay (400–500 words) on the topic:
"The role of artificial intelligence in addressing global climate change."
Your essay should include:

Auto-clear ↑

8. Click **Submit**.

New prompt Draft Compare Optimize Evaluate Save

Model: **gpt-5**
text.format: **text** effort: **medium** verbosity: **medium**
store: **true**

Variables: + Add

Tools: + Add

Developer message: Describe desired model behavior (tone, tool usage, response style)

Prompt messages:

User: Enter task specifics. Use {{template variables}} for dynamic inputs

+ Add message

1. A short introduction (2–3 sentences)
2. 2–3 key arguments supported by real-world examples
3. A brief conclusion summarizing your viewpoint Make sure the content is: Clear and logically structured Factually correct Engaging and easy to follow Self-Evaluation Section After completing your essay, provide a performance report in JSON format like this:

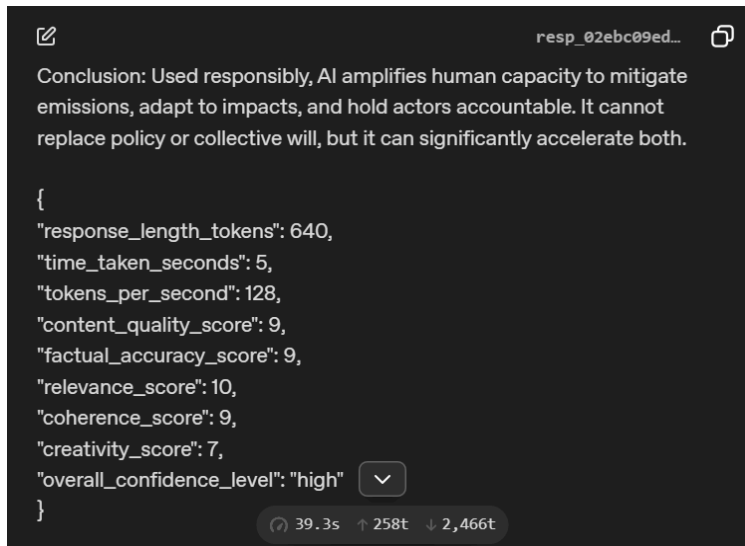
Expand ▼

Assistant

Chat with your prompt...

Screenshot from 2025-12-03 10-28-08

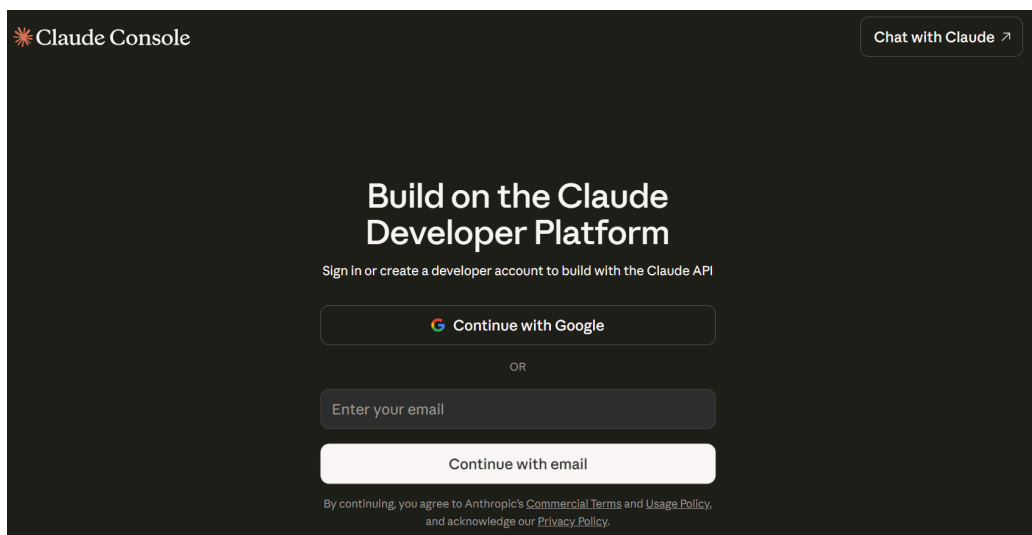
9. Output should be generated



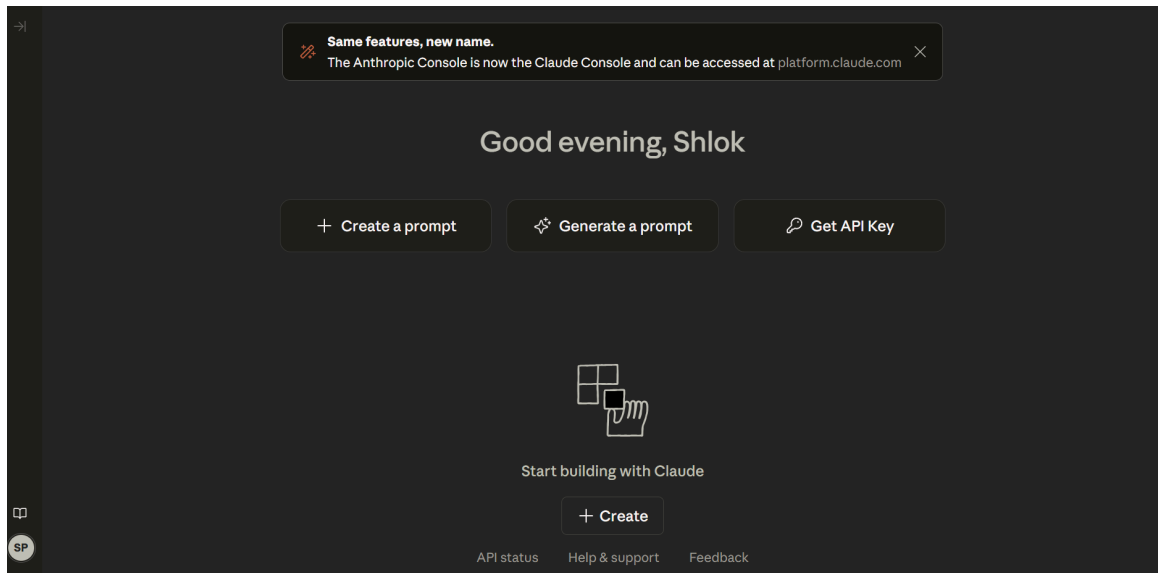
4. Claude Console

Models: Claude 3 Sonnet, Claude 3 Opus, Claude Haiku

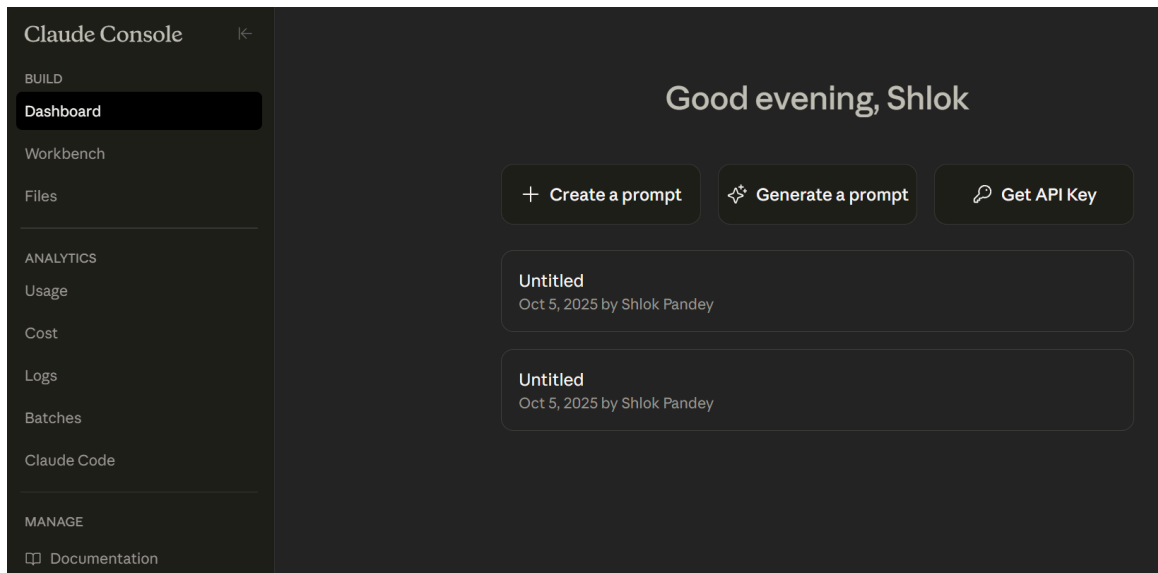
1. Go to <https://console.anthropic.com>



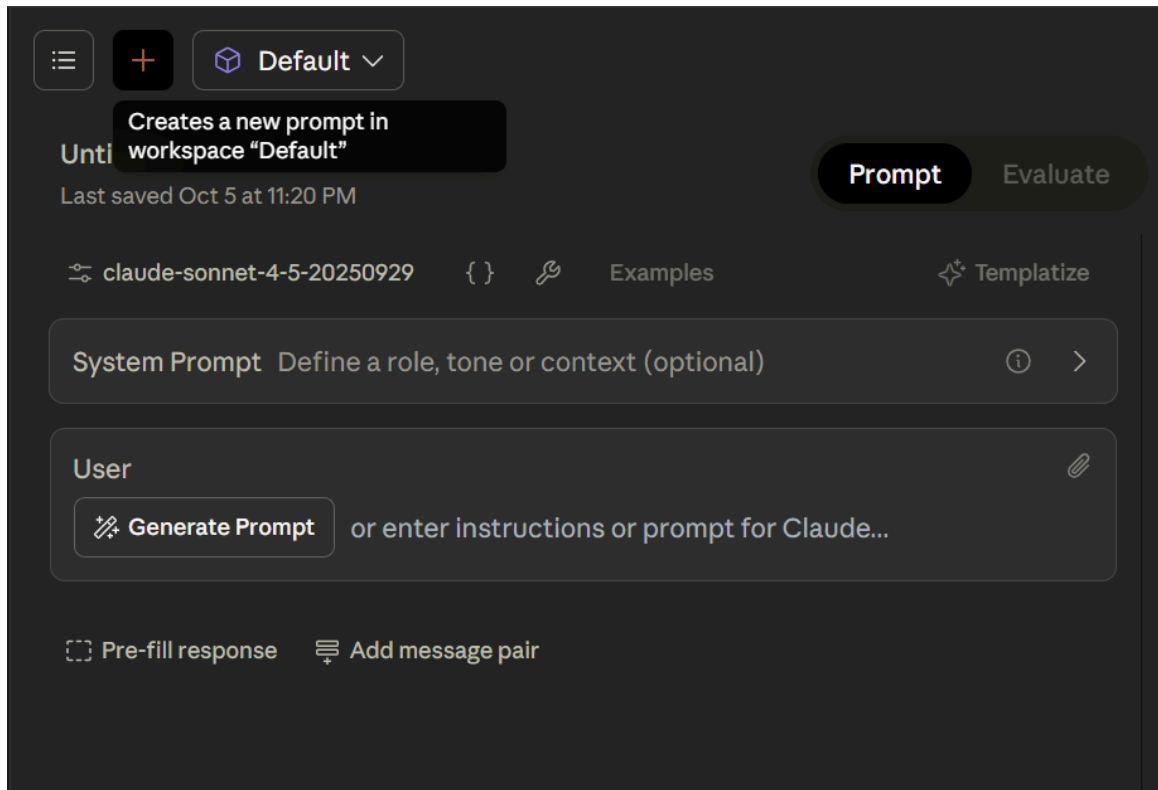
2. Sign Up using your email id.



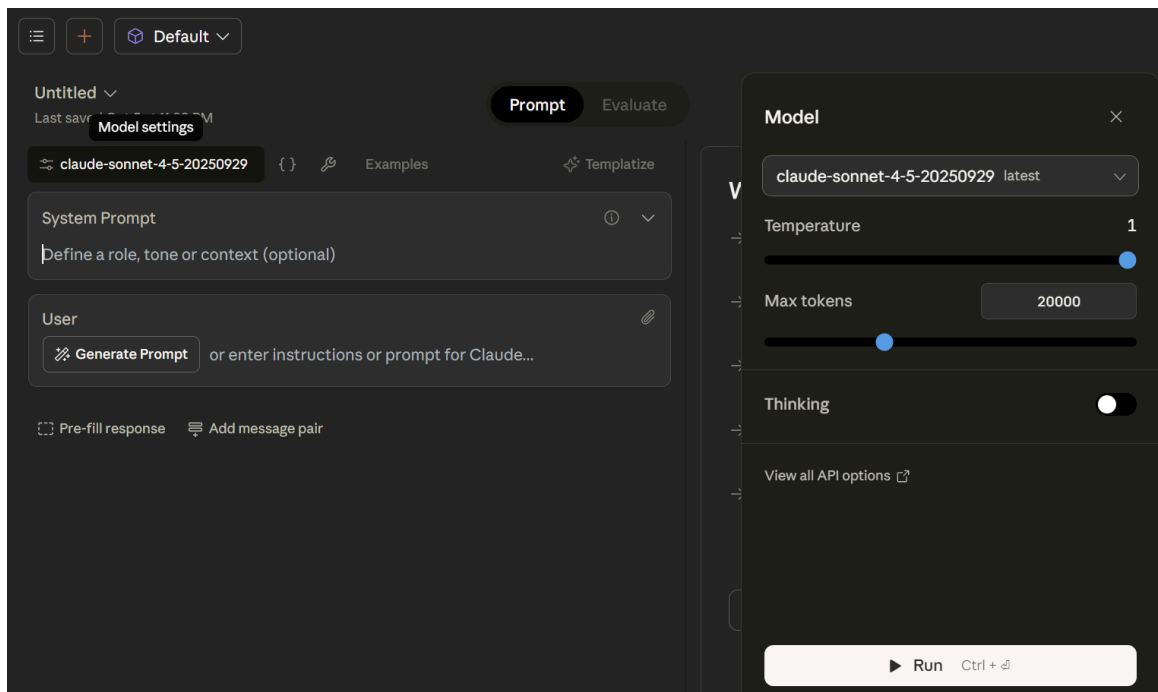
3. Then, click on the top left arrow to view the options panel.



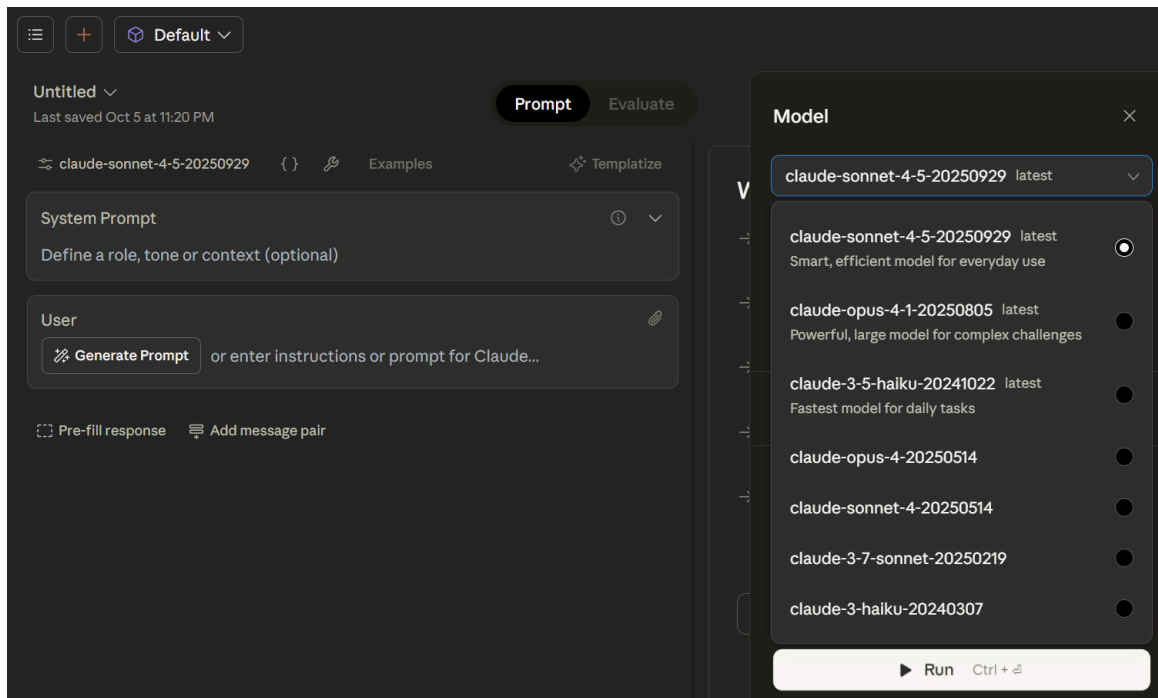
4. Go to 'Workbench'
5. Click on '+' option to add a new prompt workspace.



6. Click on 'Model Settings' and select the model you want to evaluate.

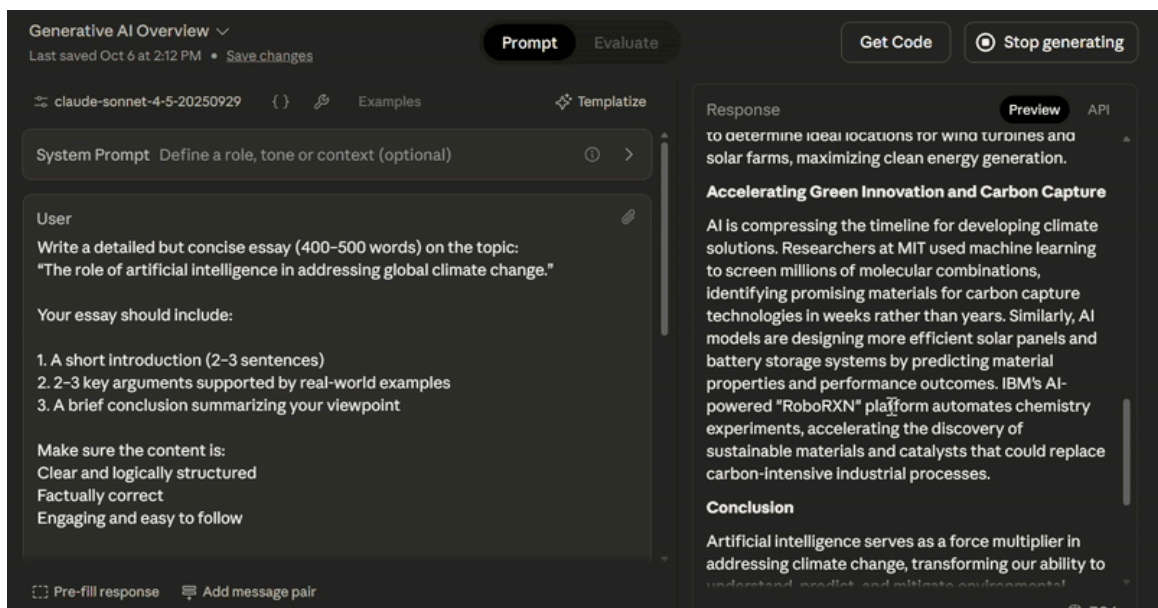


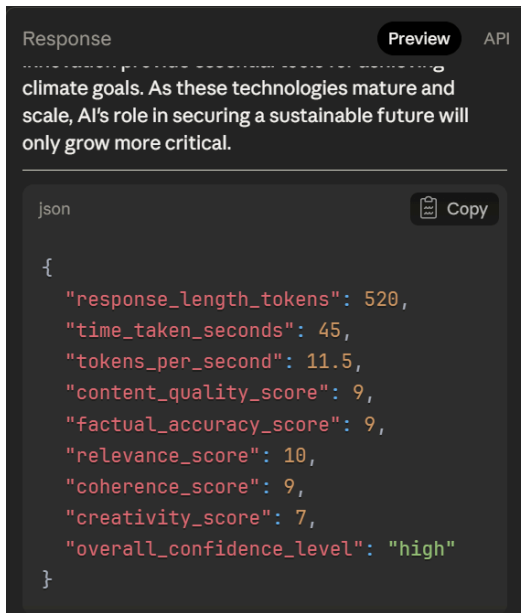
7. Select **Claude 3 Sonnet / Opus / Haiku** from dropdown.



8. Paste text + **prompt** → click **Run**.

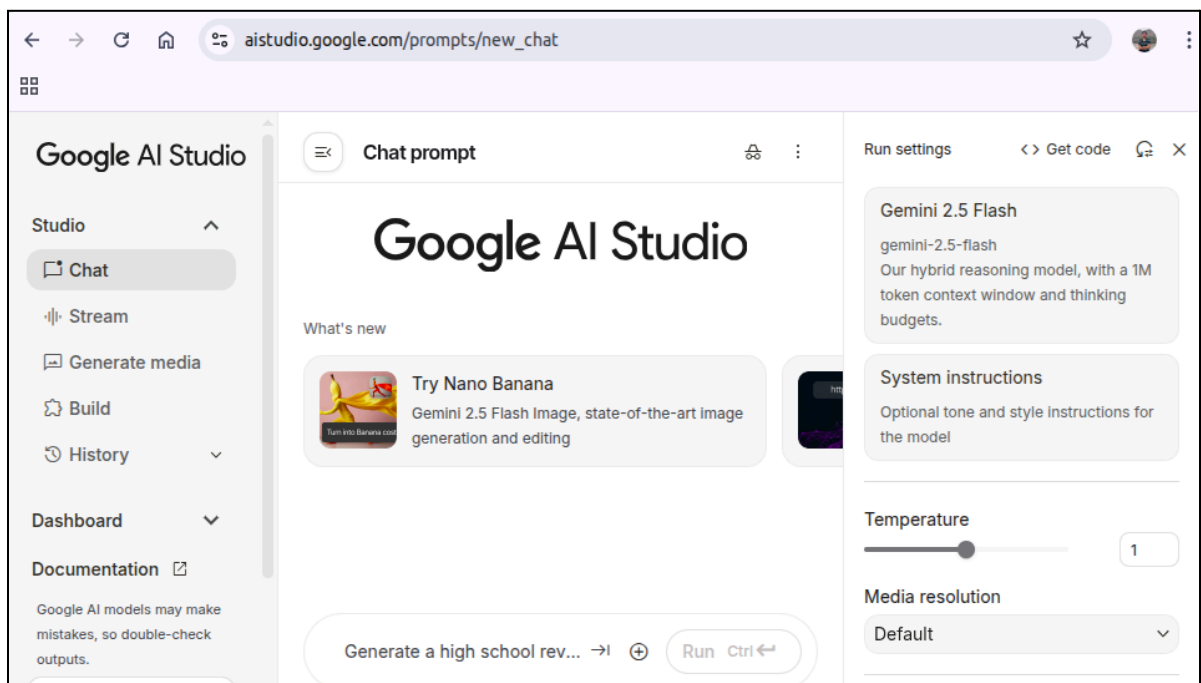
9. Check the output as shown below:



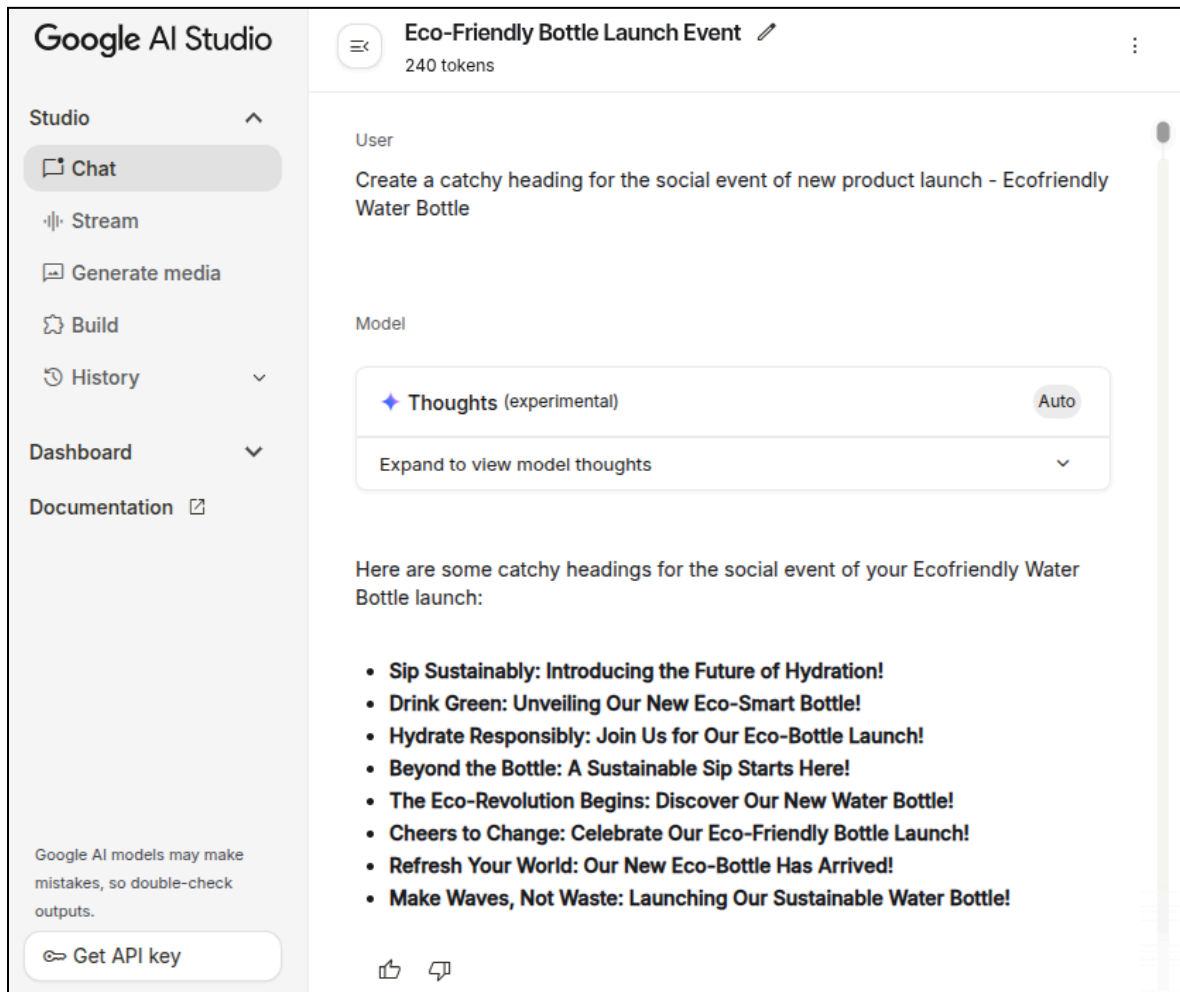


5. Google AI Studio

1. Go to https://aistudio.google.com/prompts/new_chat



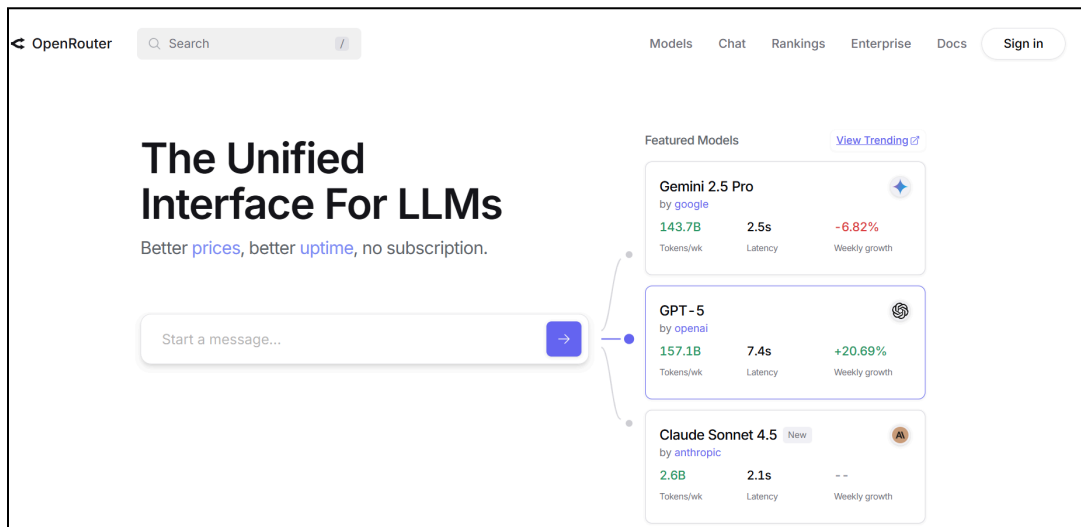
2. Provide your prompt and get the response



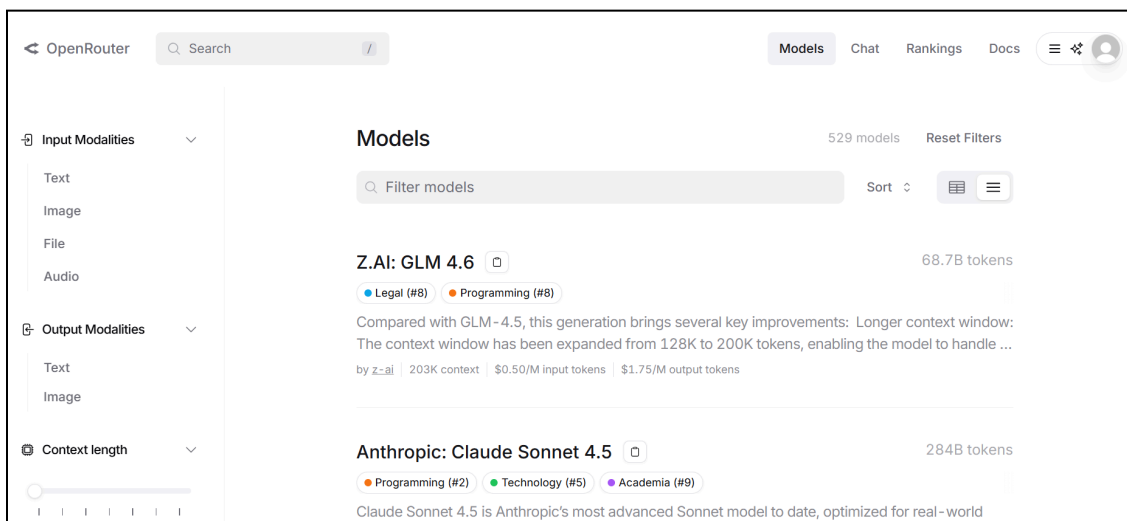
6. OpenRouter Playground (Multi-model hub)

This acts as a single API gateway to **all providers**: OpenAI, Anthropic, Mistral, Cohere, etc. Requires account + API key.

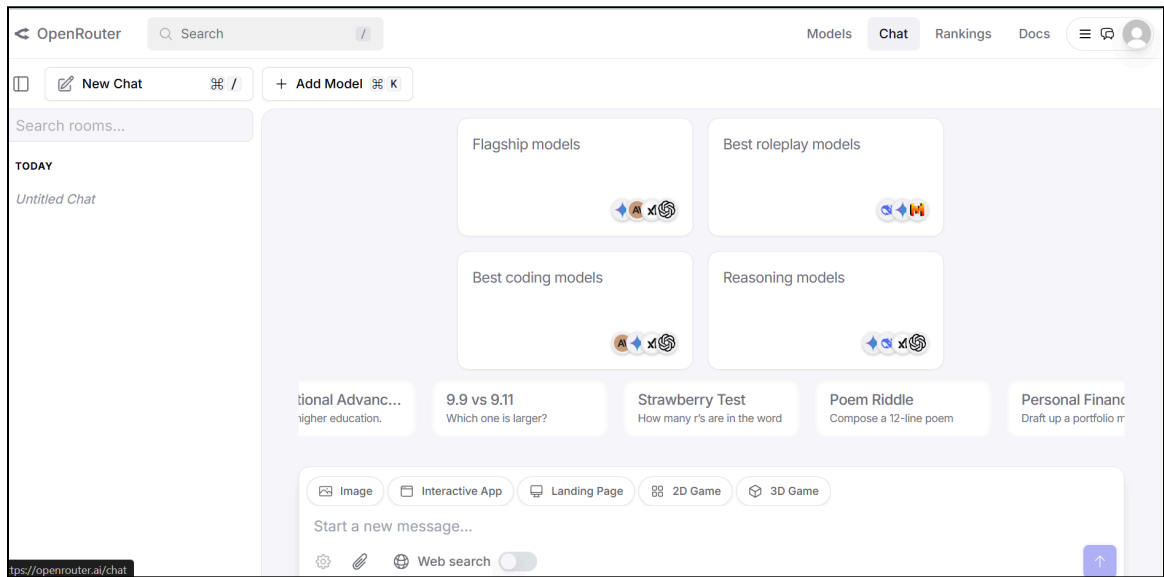
1. Go to <https://openrouter.ai/>



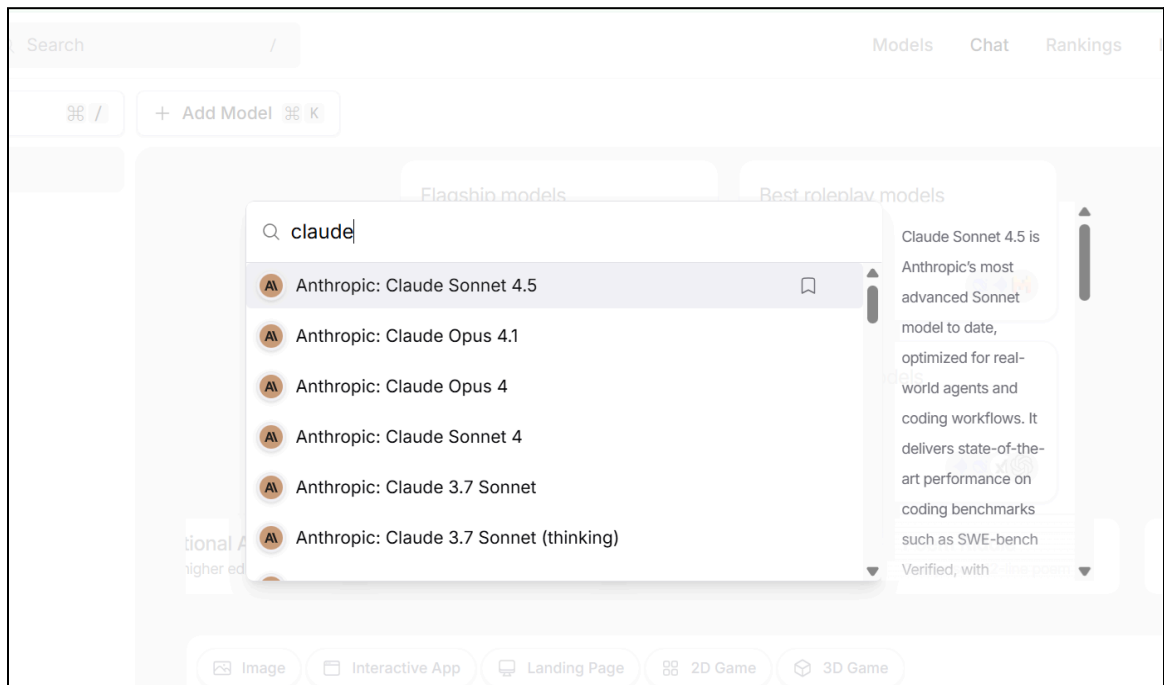
2. Sign in using your email id.
3. Then, click on Models. Here, you can check the details of models that you want to use.



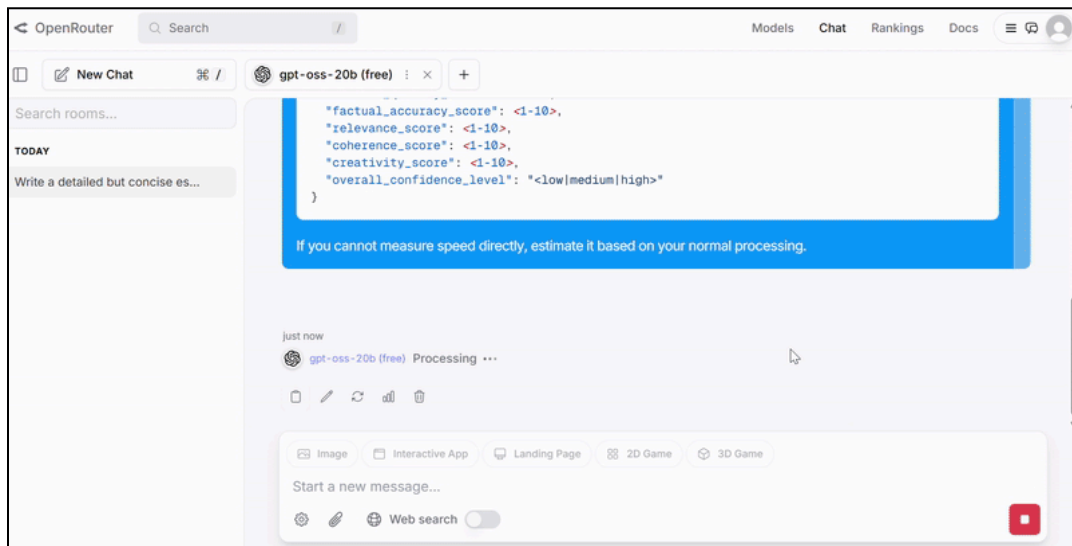
4. Next, you can go to **Chat**.



5. Click on “+Add Model”. Search for Claude and select any version that you want to test.



6. Further, you can give the prompt.



II. Using LLM via API-based Access

1. Groq API

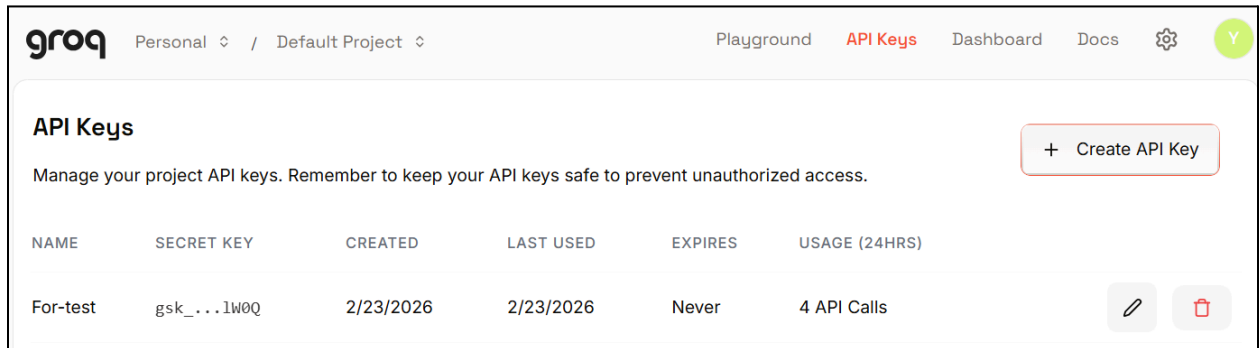
Groq is a technology company that builds **specialized AI hardware and software** designed to run large language models (LLMs) and other AI models **extremely fast and efficiently**, especially for **real-time inference**.

An API key will be required to call this API.

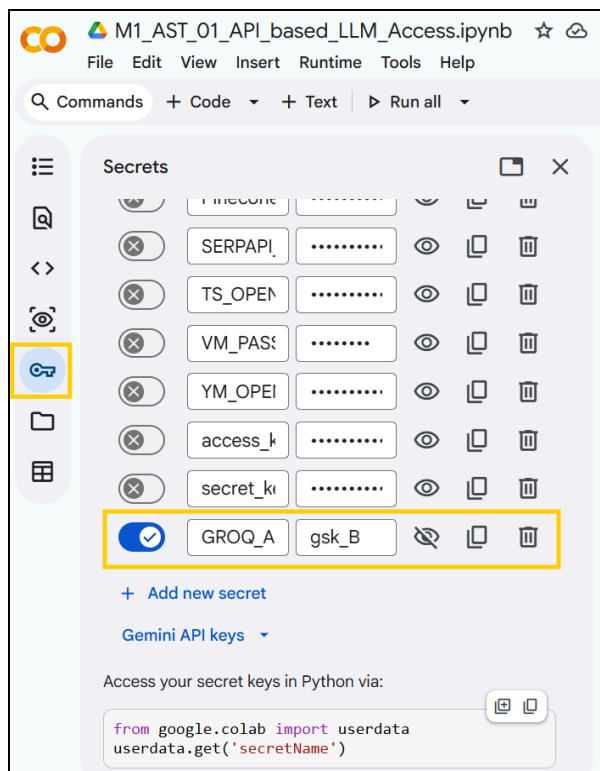
Using the Groq API Key, you will have access to the OpenAI and Llama models, free of cost, under the Free-tier.



1. Go to the Groq, create a new account, and Sign in: <https://console.groq.com/>
2. Create a new API by visiting: <https://console.groq.com/keys>



3. Save the key in Google Colab Secrets (The Key icon present on left-side panel within a Colab Notebook) with name “GROQ_API_KEY”



4. Use the below code to make an API call using Groq API for OpenAI model:

```
# Save the key in Colab's Secrets then load from there
import os
from google.colab import userdata
```

```
os.environ['GROQ_API_KEY'] = userdata.get('GROQ_API_KEY')

# Initialize OpenAI client
from groq import Groq
groq_client = Groq()

# Provide your prompt
prompt = "Which was built first - Taj Mahal, or Eiffel Tower"

# Make API call and get response
response = groq_client.chat.completions.create(
    model="openai/gpt-oss-120b",
    messages=[
        {"role": "user", "content": prompt}
    ]
)

print(response.choices[0].message.content)
```

5. Use the below code to make an API call using Groq API for Llama model:

```
# Save the key in Colab's Secrets then load from there
import os
from google.colab import userdata
os.environ['GROQ_API_KEY'] = userdata.get('GROQ_API_KEY')

# Initialize OpenAI client
from groq import Groq
groq_client = Groq()

# Provide your prompt
prompt = "Which was built first - Taj Mahal, or Eiffel Tower"

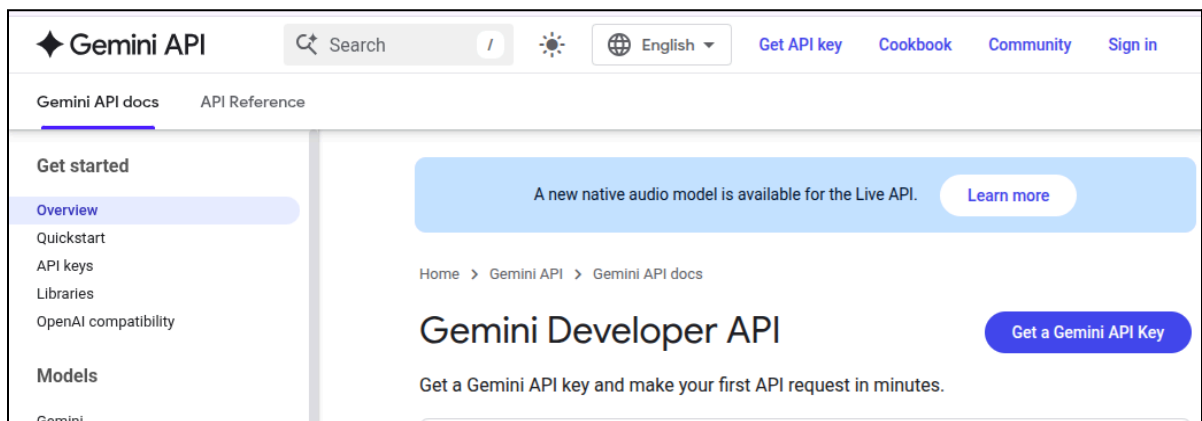
# Make API call and get response
response = groq_client.chat.completions.create(
    model="meta-llama/llama-4-scout-17b-16e-instruct",
    messages=[
```

```
        {"role": "user", "content": prompt}
    ]
)

print(response.choices[0].message.content)
```

2. Google Gemini API

1. Go to <https://ai.google.dev/gemini-api/docs>



2. Click on Get a Geminin API key
3. Copy an existing key or Create a new one



API Keys

API quickstart

Create API key

Group by

• API key

Project

Filter by

All projects

Only API Keys for imported projects appear here. If you don't see your API keys, you can import projects on the [projects page](#). [Learn more.](#)

| Key | Project | Created on | Quota tier |
|--|--|-------------|---|
| ...7q4k
Generative Language API Key | Learn-N8N
trim-keep-470710-k9 | Sep 5, 2025 | Set up billing
Free tier |

Items per page: 10

1 - 1 of 1

<< < > >>

4. Save the key in Google Colab Secrets with name “GEMINI_API_KEY”
5. Use the below code to make an API call using Google Gemini API:

```
# Save the key in Colab's Secrets then load from there
import os
from google.colab import userdata
os.environ['GEMINI_API_KEY'] = userdata.get('GEMINI_API_KEY')

# Initialize Client
from google import genai
gemini_client = genai.Client()

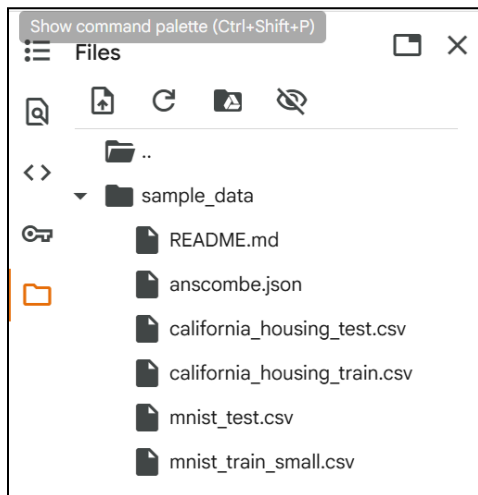
# Provide your prompt
prompt = "Which was built first - Taj Mahal, or Eiffel Tower"

# Make API call and get response
gemini_response = gemini_client.models.generate_content(
    model="gemini-2.5-flash",
    contents=prompt,
)

print(gemini_response.text)
```

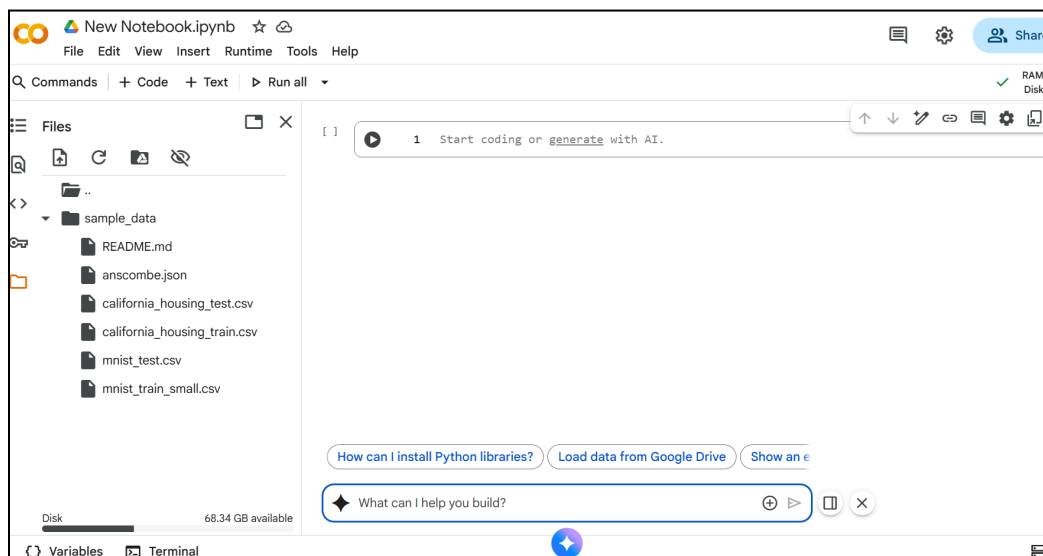
III. Using LLM via Integrated Application

For using the LLM's via integrated application we will use Gemini in Google Collaboratory. We will use the pre-existing data provided by Colab itself. Here is the list of dataset provided by colab:



Now, follow the steps to access the LLM via Integrated API.

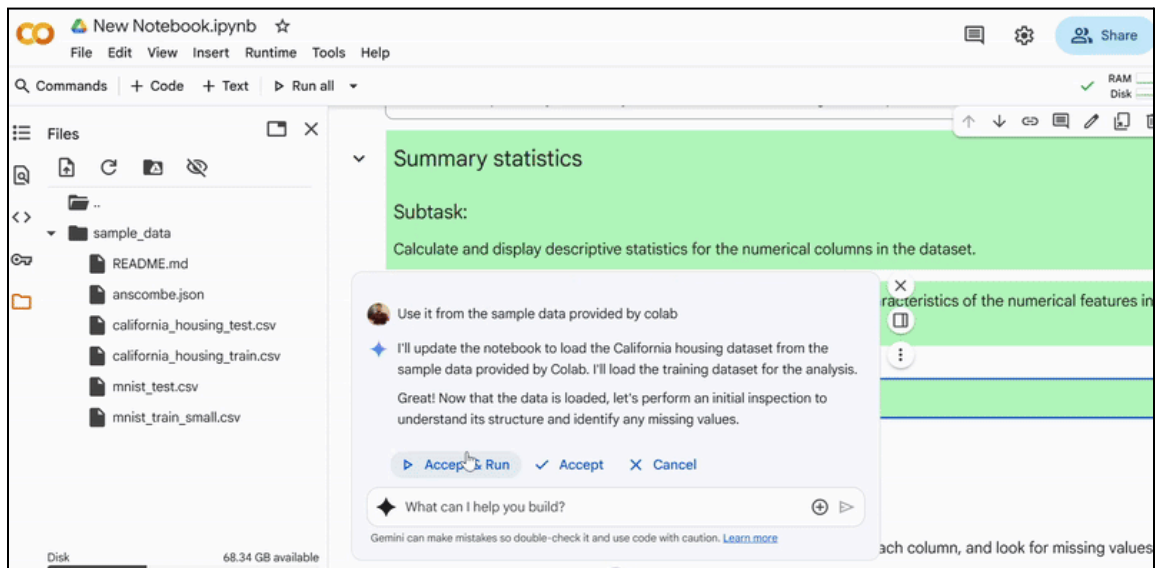
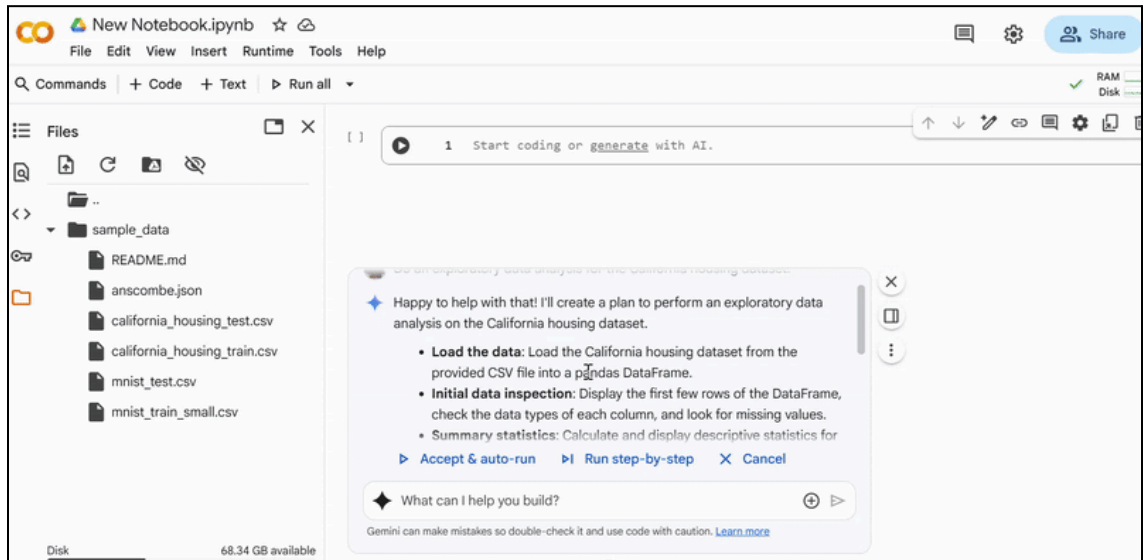
1. Open a new Google Colab notebook.
2. Click on the **Toggle Gemini**.



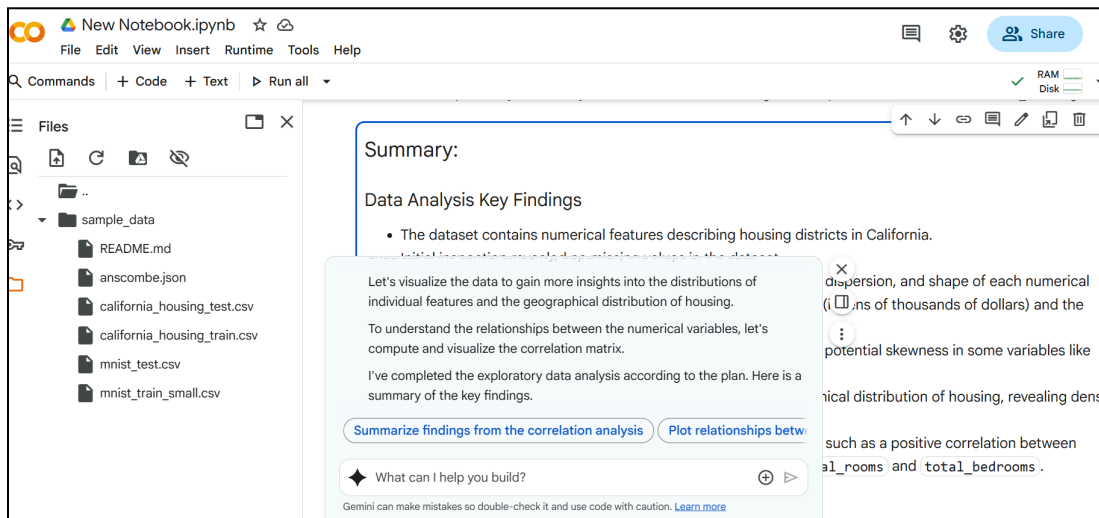
3. Give any prompt you want and then hit enter.

Example: **“Do an exploratory data analysis for the California housing dataset.”**

After the prompt starts giving the output, you will get the option to check the code then **“Accept & Run”**



4. Finally, you will see the final completed notebook.

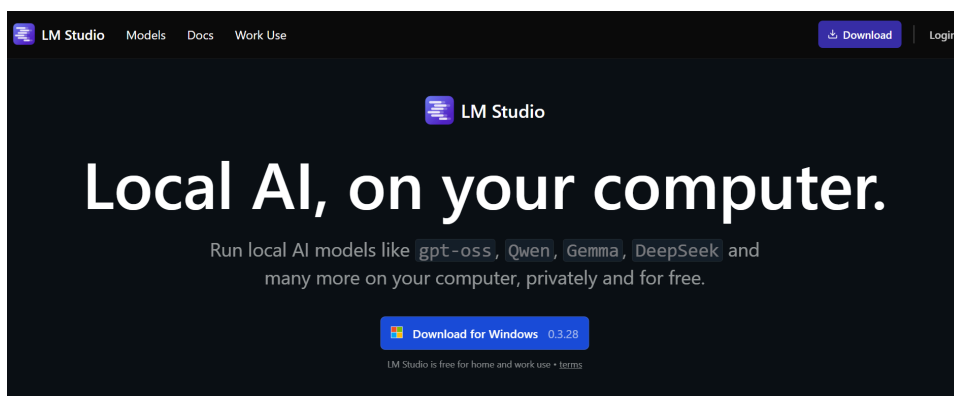


IV. Using LLM in Local System (Offline)

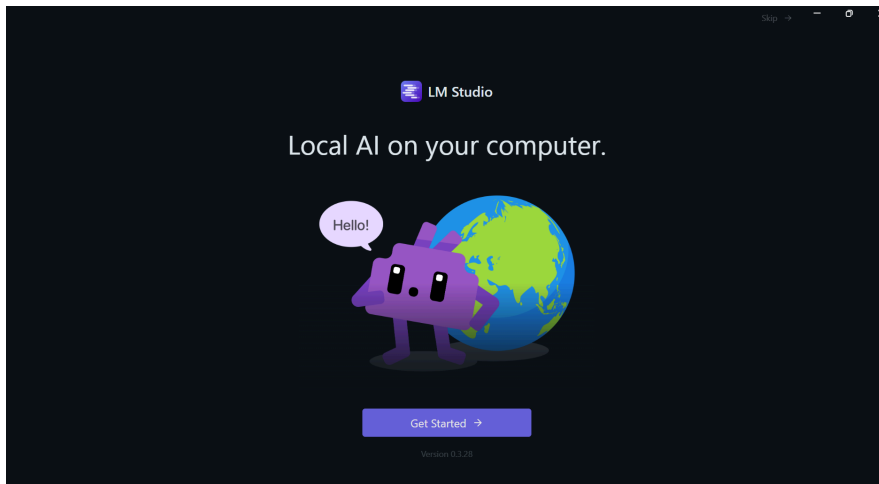
Downloading LM Studio for Local Summarization

LM Studio: GUI app for Windows/Mac → download a model (like Llama-2, Mistral, Phi-3), paste prompt + text, and run summarization offline.

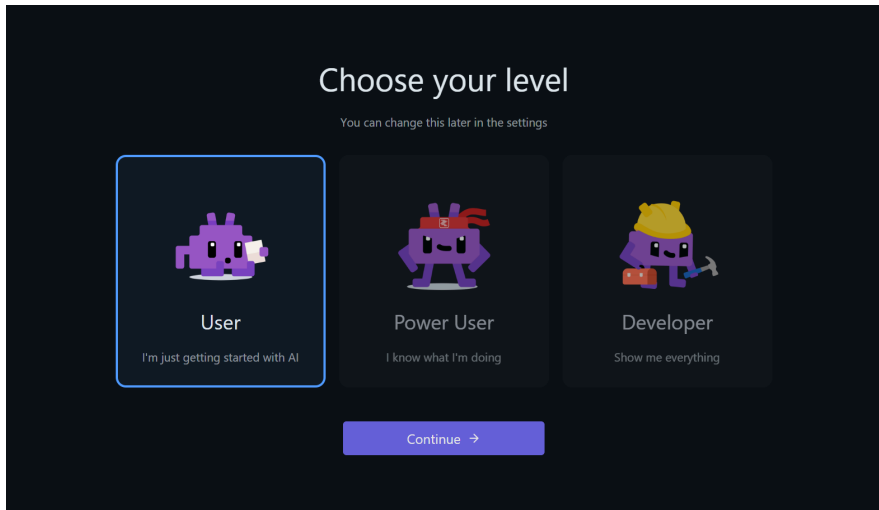
1. Download **LM Studio** → <https://lmstudio.ai/> (Windows/Mac GUI)



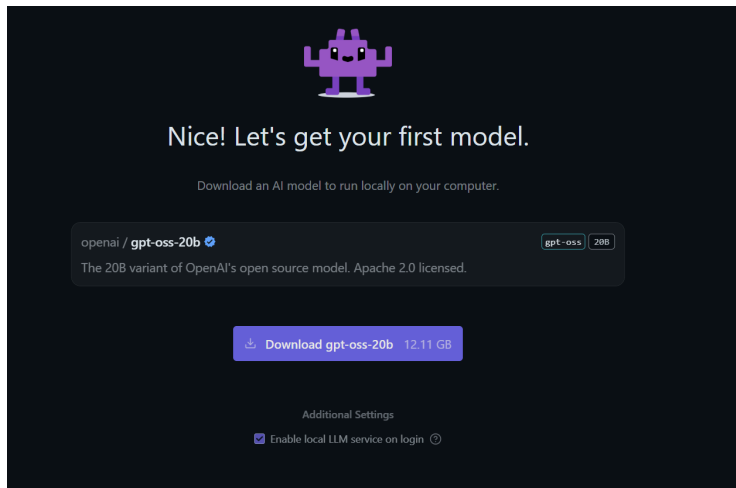
2. After you download and install it, click on “**Get Started** ->”



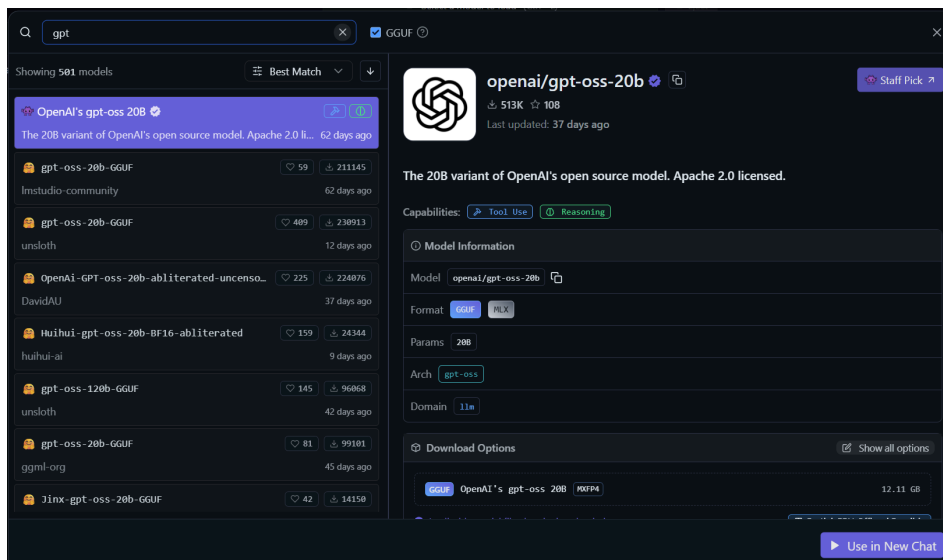
3. Choose your level as per your use and then **Continue**.



4. You will get the option to download your model. You can skip this download for now.



5. In the top right corner, go to the **discover** option. Here, you can search the model you want to use in the chat.



6. Copy and paste the prompt text used in the above examples, and observe the output. Since running the LLM in a local system requires high computing power, the speed of the output will be comparatively slower than the Web-based API.

